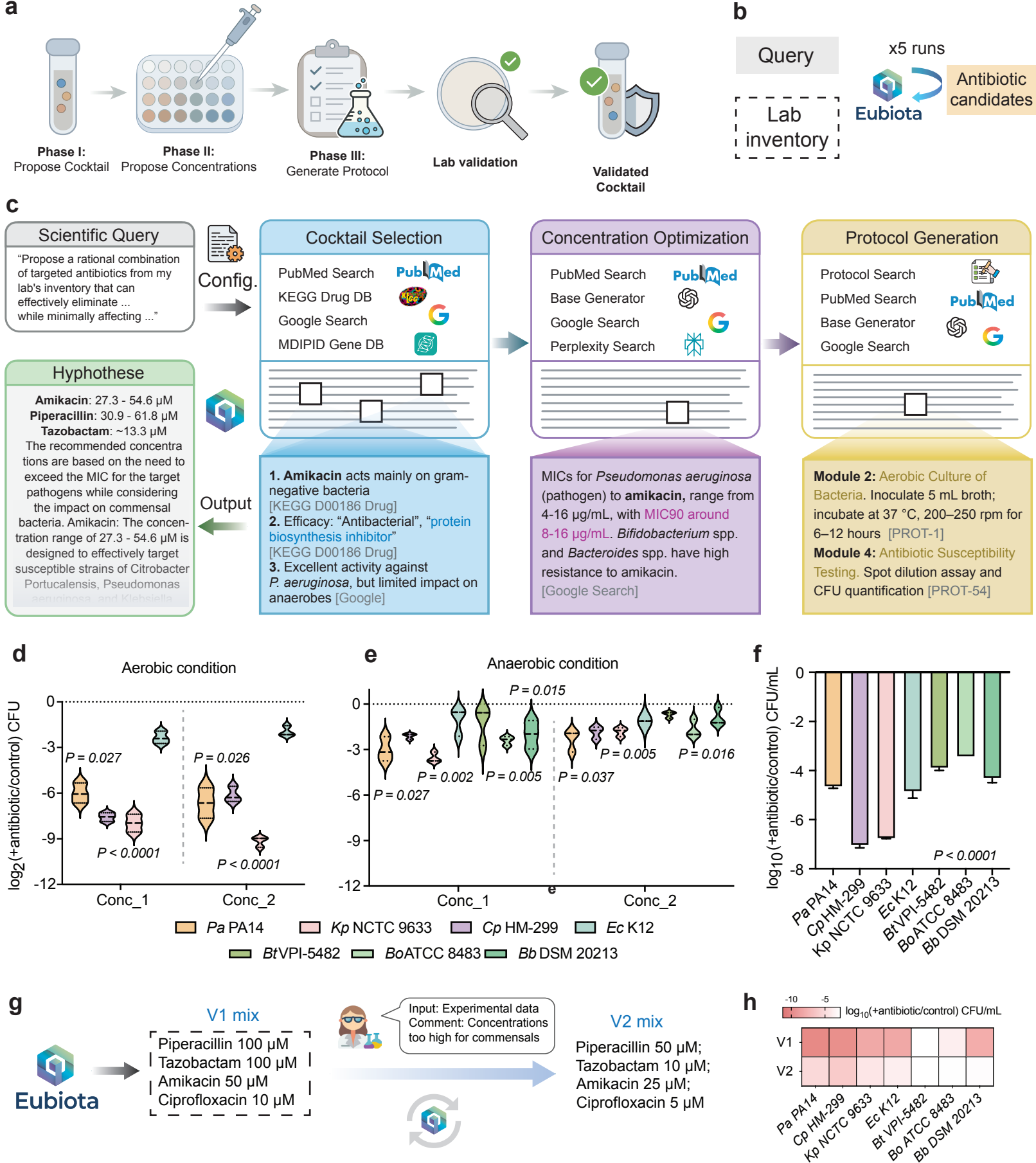
a
Phase I:
Propose Cocktail
Phase II:
Propose Concentrations
Phase III:
Generate Protocol
Lab validation
Validated Cocktail
b
Query
Lab inventory
x5 runs
Eubiota
Antibiotic candidates
c
Scientific Query
"Propose a rational combination of targeted antibiotics from my lab's inventory that can effectively eliminate ... while minimally affecting ..."
Config.
Hyphothese
Amikacin: 27.3 - 54.6 μM
Piperacillin: 30.9 - 61.8 μM
Tazobactam: ~13.3 μM
The recommended concentrations are based on the need to exceed the MIC for the target pathogens while considering the impact on commensal bacteria. Amikacin: The concentration range of 27.3 - 54.6 μM is designed to effectively target susceptible strains of Citrobacter Portucalensis, Pseudomonas aeruginosa, and Klebsiella
Output
Cocktail Selection
PubMed Search
KEGG Drug DB
Google Search
MDIPID Gene DB
1. Amikacin acts mainly on gram-negative bacteria [KEGG D00186 Drug]
2. Efficacy: "Antibacterial", "protein biosynthesis inhibitor" [KEGG D00186 Drug]
3. Excellent activity against P. aeruginosa, but limited impact on anaerobes [Google]
Concentration Optimization
PubMed Search
Base Generator
Google Search
Perplexity Search
MICs for Pseudomonas aeruginosa (pathogen) to amikacin, range from 4-16 μg/mL, with MIC90 around 8-16 μg/mL. Bifidobacterium spp. and Bacteroides spp. have high resistance to amikacin. [Google Search]
Protocol Generation
Protocol Search
PubMed Search
Base Generator
Google Search
Module 2: Aerobic Culture of Bacteria. Inoculate 5 mL broth; incubate at 37 °C, 200–250 rpm for 6–12 hours [PROT-1]
Module 4: Antibiotic Susceptibility Testing. Spot dilution assay and CFU quantification [PROT-54]
d
Aerobic condition
log2(+antibiotic/control) CFU
P = 0.027
P < 0.0001
P = 0.026
P < 0.0001
Conc_1
Conc_2
e
Anaerobic condition
P = 0.015
P = 0.027
P = 0.002
P = 0.005
P = 0.037
P = 0.005
P = 0.016
Conc_1
Conc_2
Pa PA14
Kp NCTC 9633
Cp HM-299
Ec K12
Bt VPI-5482
Bo ATCC 8483
Bb DSM 20213
f
log10(+antibiotic/control) CFU/mL
P < 0.0001
Pa PA14
Cp HM-299
Kp NCTC 9633
Ec K12
Bt VPI-5482
Bo ATCC 8483
Bb DSM 20213
g
Eubiota
V1 mix
Piperacillin 100 μM
Tazobactam 100 μM
Amikacin 50 μM
Ciprofloxacin 10 μM
Input: Experimental data
Comment: Concentrations too high for commensals
V2 mix
Piperacillin 50 μM;
Tazobactam 10 μM;
Amikacin 25 μM;
Ciprofloxacin 5 μM
h
log10(+antibiotic/control) CFU/mL
V1
V2
Pa PA14
Cp HM-299
Kp NCTC 9633
Ec K12
Bt VPI-5482
Bo ATCC 8483
Bb DSM 20213